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Factors influencing pedestrians' decision to cross the road by risky rolling gap crossing strategy at intersections in Dhaka, Bangladesh

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ABSTRACT

Pedestrian road-crossing strategy is one of the most important pedestrian road-crossing behaviors. The safety of the pedestrians often depends on it. Among the road-crossing strategies, rolling gap crossing strategy is the riskiest one. The objective of this research was to explore the factors that influenced pedestrians' decision to cross the road by rolling gap crossing at intersection. Data regarding road-crossing strategy of the pedestrians, their characteristics, their road-crossing behavior, intersection geometry, and traffic environmental condition were collected through videography survey method, on-site observation, and secondary source from six intersections of Dhaka, Bangladesh. A binary logistic regression model was developed in this study by using the collected data. Results of the developed model showed that seven statistically significant factors strongly influenced pedestrians' decision to cross the road by rolling gap crossing at intersections. These factors were intersection control type, median width, vehicle flow, available gap on the road, age group of the pedestrians, their crossing group size, and their behavior of crosswalk usage. The results of this study would help the policymakers to take proper interventions to alleviate pedestrian safety problems.

1. Background of the study

Traffic crashes involving pedestrians have become a major safety concern throughout the world, particularly in the developing countries. About 1.35 million people died each year due to traffic crashes that were the eighth leading cause of death around the world. About 22% of the total deaths were pedestrians and 90% of the total deaths occurred in the developing countries (World Health Organization (WHO), 2018). In Bangladesh, the situation is more severe as pedestrians are the most neglected road-user group. According to Accident Research Institute (ARI) of Bangladesh University of Engineering and Technology (BUET), about 28,244 pedestrian casualties occurred from 1998 to 2013 in Bangladesh that represented 34% of all the traffic-related casualties (Road Safety Facts, 2013). Pedestrian safety scenario was found in the worst condition in Dhaka, the capital city of Bangladesh. More than 10 thousand crashes occurred in Dhaka from the year 1998 to 2014 and 4,514 pedestrians died in those crashes (Road Safety Facts, 2014). Therefore, it could be concluded that pedestrian safety has become a major concern that needs to be addressed properly.

Pedestrian road-crossing behavior is an important issue that describes how pedestrians value their time, their lives, how they perceive their surrounding environment, how their behavior changes during crossing, and how they apply different tactics for crossing the road (Kadali and Perumal, 2012). Pedestrians are one of the most interactive decision-makers and their road-crossing decisions are unpredictable as they show diverse road-crossing behaviors which could vary from one situation to another (Amini et al., 2019). Therefore, factors influencing pedestrians' road-crossing decisions to choose crossing behaviors need to be explored to improve their safety on the road (Zhang et al., 2018, 2019). In research studies on pedestrian safety, pedestrians' crossing decision-making process is mostly considered in terms of their violation behavior (Rosenbloom, 2009; Zafri et al., 2019; Koh et al., 2014; Tom and Granié, 2011; Guo et al., 2011; Brosseau et al., 2013; Zhuang et al., 2018; Cherry et al., 2012) and gap acceptance behavior (Yannis et al., 2013; Kadali and Perumal, 2012; Shaaban et al., 2019; Chandra et al., 2014; Koh and Wong, 2014). In violation behavior, researchers examined either a pedestrian complied with rules (e.g., comply with traffic signals, comply with traffic police direction, comply with crosswalk) or not. In gap acceptance behavior, researchers explored either a pedestrian accepted the available time (or space) gap between the pedestrian and vehicle or not. Some researchers also conducted similar studies that focused on factors influencing the accepted gap size

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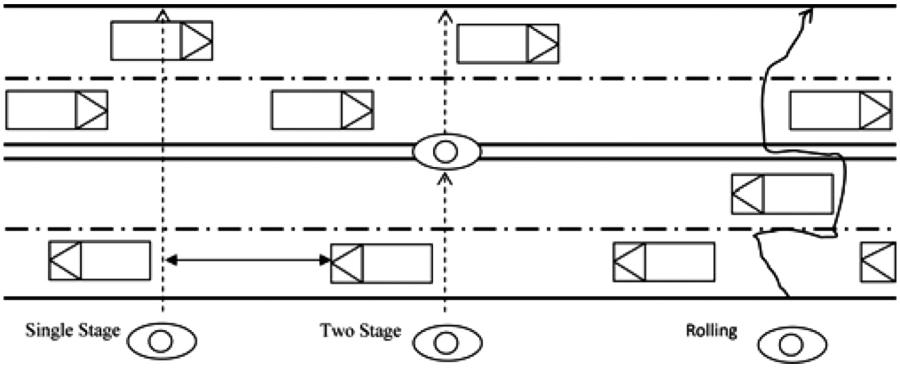


Fig. 1. Pedestrian road-crossing strategies (Chandra et al., 2014).

(e.g., critical gap size, average gap size, minimum gap size) of the pedestrian.

Pedestrian road-crossing strategy is one of the most important pedestrian road-crossing behaviors (Zhang et al., 2019). Brewer et al. (2006) and Zhang et al. (2018) simplified the pedestrian crossing strategy into three types: single-stage, two-stage, and rolling gap (see Fig. 1). According to them, in single-stage crossing, pedestrians continuously cross the road in one crossing maneuver irrespective of road width. In addition, pedestrians do not roll over the small vehicular gap and do not stop in the median while crossing the road by single-stage crossing. In two-stage crossing, pedestrians cross the road up to the median in one go then wait a while, and subsequently cross the far-side. Although both of the crossing strategies are almost the same, there is a subtle difference between them. In two-stage crossing, pedestrians have to wait on the median for finding appropriate gaps to cross the far-side of the road; while in single-stage crossing, they do not need to wait on the median. The last crossing strategy is rolling gap crossing where pedestrians keep searching for appropriate gaps between the continuous flow of vehicles by adjusting their speed and direction of movement, and consequently they roll over the small vehicular gaps to cross the road. They cross the road discontinuously and may stop or adjust their speed to avoid conflict with the vehicles.

Pedestrians often make crossing related decisions based on available gaps on the road (Chandra et al., 2014; Cherry et al., 2012; Kadali and Vedagiri, 2013). They do not always want to wait, particularly roads with high traffic volume, to cross the road when all lanes are completely clear; therefore, they rely on the rolling gap crossing strategy (Brewer et al., 2006; Kadali and Vedagiri, 2013). In this crossing, they anticipate that the lanes would be cleared when they cross. Therefore, they may start crossing the road after finding a proper gap in the nearest lane. Pedestrians may stop in the middle of the road after one successful gap acceptance and look for another suitable gap in the next lane. Otherwise, they may cross the lane immediately as it has been cleared after their first lane crossing (Arman et al., 2015). Since they try to move through the available gaps while crossing the road by rolling gap crossing, sometimes they may accept small gap. Accepting small gap results in pedestrian-vehicle conflict which increases the crash risk of pedestrians (Kadali and Perumal, 2012). Previous studies identified that among the crossing strategies, rolling gap crossing is the riskiest one followed by two-stage and single-stage crossing (Brewer et al., 2006; Zhang et al., 2019). Koh and Wong (2014) advised the pedestrians not to cross the road by risky rolling gap crossing strategy. They showed that there is an excessive chance for pedestrians to be injured during the rolling gap crossing (60% at near end crossing and 77% at far end crossing).

Indeed, whilst a large number of researches tried to understand the decision-making process of pedestrians in terms of their violation behavior or gap acceptance behavior, researchers provided negligible attention to explore the rolling gap crossing and factors that influenced pedestrians' decision to choose this crossing strategy to cross the road. Only a study of Zhang et al. (2018) was found where they put focus on it. They developed a multinomial logistic model to examine the relationship between pedestrian crossing strategy and factors, including gender, age, waiting time, and traffic volume at uncontrolled multi-lane mid-block crosswalk in China. They found that male and younger pedestrians showed fewer rolling gap crossing than female and older pedestrians, respectively. While the chance of showing rolling gap crossing increased with increasing traffic volume and waiting time. However, the authors of this study claimed that the accuracy of the above-mentioned results was not satisfactory due to lack of sample size as well as poor accuracy of the model for ignoring important factors mainly related to pedestrian road-crossing behaviors in the model development process. They addressed these limitations by developing pedestrian dynamic gap acceptance (PDGA) model to describe pedestrian road-crossing behavior and their decision-making procedure. Therefore, it could be said that there is still a lack of literature on pedestrian rolling gap crossing. Hence, this study aimed to explore pedestrian rolling gap crossing and factors that pushed pedestrians' decision to choose rolling gap crossing to cross the road at intersection irrespective of the risk associated with this type of crossing strategy.

2. Literature review

To fulfill the objective, first, it was required to identify the potential factors that should be considered to explore pedestrians' decision to cross the road by rolling gap crossing strategy. Although very few attempts were made to identify the factors that could influence pedestrian crossing strategy, literature on other pedestrians' crossing decision-making process (e.g., violation behavior, gap acceptance behavior) had offered some clues on potential factors. Therefore, an extensive literature review was carried out to identify the potential factors that influenced the violation behavior and gap acceptance behavior of the pedestrians. Those factors were used later to examine their influence on pedestrians' decision to cross the road by rolling gap crossing at intersection.

2.1. Intersection geometry and traffic environmental characteristics

Previous literature showed that intersection geometry and traffic environmental condition influenced pedestrians' decision-making process. In case of intersection geometry attributes, some studies indicated that violation behavior of pedestrians was influenced by intersection control type. Asaithambi et al. (2016) demonstrated that both violation behavior and accepted gap size reduced after installing signal at intersection. Brosseau et al. (2013) showed that pedestrians were less likely to show violation behavior at intersections where pedestrian signals with countdown displays were present. In addition, Zafri et al. (2019) mentioned that pedestrians were found to violate more at intersection which was controlled by traffic police based on their own judgment than that intersection which was controlled by traffic police

Table 1
Details of the selected intersections.

Intersection name	Control type	Types of intersection	Number of legs	Condition of crosswalk	Focused leg width (m)	Focused leg median width (m)	Collected sample
Gulshan-2	Controlled	Cross	4	Highly visible	26	1.5	109
Bangla Motor	Controlled	Cross	4	Highly visible	30	1.6	71
Shyamoli	Uncontrolled	Tee	3	Not available	31	1.2	100
Sonargaon-Panthapth-ETV	Controlled	Roundabout	5	Poorly visible	29	3.0	86
Shapla Chattar	Controlled	Roundabout	4	Not available	28	1.0	85
Jasim Uddin	Controlled	Tee	3	Poorly visible	24	2.0	95

implementing the direction of vehicular traffic signals. Width of the road, availability and width of the median (refuge island), as well as crosswalk condition and its visibility also influenced pedestrians' decision. On narrow roads, pedestrians were more likely to violate (Koh et al., 2014) and accept shorter gap size (Schmidt and Faerber, 2009). Moreover, presence of median (refuge islands) decreased the compliance rate of pedestrians (Koh et al., 2014); in contrast, opposite result was found for the presence of high visibility markings (Turner et al., 2006). Furthermore, pedestrians were more likely to accept a smaller gap while waiting on the median than the curbside (Hamed, 2001).

In case of traffic environmental attributes, several researchers showed that available gap on the road and traffic volume (vehicle flow) influenced pedestrians' decision. The likelihood of pedestrian gap acceptance was found higher when larger gap was available on the road (Koh and Wong, 2014; Brewer et al., 2006; Kadali and Vedagiri, 2013; Chandra et al., 2014). Furthermore, Chandra et al. (2014) demonstrated that accepted gap size decreased with an increase in the traffic volume. In addition, pedestrians were more likely to show violation behavior with an increase in the number of passing vehicles (Zafri et al., 2019; Koh et al., 2014).

2.2. Pedestrian characteristics

Many studies tried to understand the relation between pedestrians' decision-making process and their characteristics. Among the pedestrian characteristics, some studies showed that gender of the pedestrians influenced their crossing decisions. They showed that male pedestrians were found to violate traffic rules more than female pedestrians (Lipovac et al., 2013; Tom and Granié, 2011; Rosenbloom, 2009, 2011; Guo et al., 2011). Aligned with that, male pedestrians were also found to accept smaller gap compared to female pedestrians (Kadali and Perumal, 2012; Nor et al., 2017). Both of the results indicated higher risk taking tendency of the male pedestrians. Along with gender, age of the pedestrians was also found one of the factors influencing pedestrians' decision. Younger pedestrians were more likely to violate rules (Hao et al., 2008; Ren et al., 2011; Granié et al., 2013; Cook and Koorey, 2013) and accept smaller gap (Beggiato et al., 2017; Sun et al., 2003; Amini et al., 2019) while crossing than older pedestrians. However, several studies also showed a different finding, where younger pedestrians accepted larger gap than older pedestrians (Dommes et al., 2013, 2014).

2.3. Pedestrian behavior

Road-crossing behaviors of pedestrians have an effect on their decision-making process. In case of group crossing behavior, previous studies showed that pedestrians who crossed the road in group were less likely to violate traffic rules (Rosenbloom, 2009; Marisamynathan and Perumal, 2014; Zhuang and Wu, 2011; Brosseau et al., 2013) and accept larger gap (Yannis et al., 2013) than that of pedestrians who crossed alone. On the other hand, the likelihood of accepting smaller gap increased with an increase in the crossing speed of the pedestrians (Chandra et al., 2014; Kadali and Perumal, 2012). Aligned with that, pedestrians showed more violation behavior (Koh et al., 2014; Li, 2013; Guo et al., 2011; Wang et al., 2011) and accepted smaller gap (Antić et al., 2016; Cherry et al., 2012) with an increase in the waiting time of the pedestrian.

3. Methodology

3.1. Study area selection

To explore the factors that influenced pedestrians' decision to cross the road by rolling gap crossing at intersections, six busy intersections of Dhaka city had been selected deliberately. If all the legs of an intersection were found as major roads of Dhaka according to Dhaka Transport Coordination Authority (DTCA), then the intersection was considered as a busy intersection. Among the intersections, there were two cross intersections, two tee intersections, and two roundabout intersections. The intersections included five controlled intersections and an uncontrolled intersection. For collecting data, one leg of each intersection was focused. Crosswalk was available in the four focused legs of the six intersections. Sidewalk and median were found in all of the focused legs of the studied intersections. Commercial land use was the most common surrounding land use of each intersection. Table 1 represents the description of the study sites; whereas, Fig. 2 presents photographs.

3.2. Data collection

Data regarding road-crossing strategy of the pedestrians, their characteristics, their road crossing behaviors, and traffic environment were collected via videography survey method under fair weather condition. A normal working day was selected for the survey to capture typical busy roadway condition. The survey was done during the morning (10.30 am − 12.30 pm) and evening (04.30 pm − 06.30 pm) peak periods. The video footages were recorded through a high resolution camera (Canon EOS 70D) which can take 60 frames per second, from an appropriate elevated point (06–10 meters above from road surface) from where the focused leg of the intersection was well covered as well as the pedestrians were clearly visible. On the other hand, data related to intersection geometry were collected from on-site observation and secondary source.

3.3. Data extraction

Details about the collected data are shown in Table 2. Data related to intersection control type and crosswalk condition were collected at the time of conducting videography survey through on-site observation. Road and median width data were measured from the shapefiles of Detailed Area Plan (DAP) of Dhaka which were prepared by Capital Development Authority (RAJUK). The rest of the necessary data were extracted manually from the video footages by following a systematic procedure. The authors prepared a guideline for extracting the required data properly. Five data extractors were trained by an author following the prepared guideline. Those data extractors extracted data independently under the supervisor of the author who trained them.

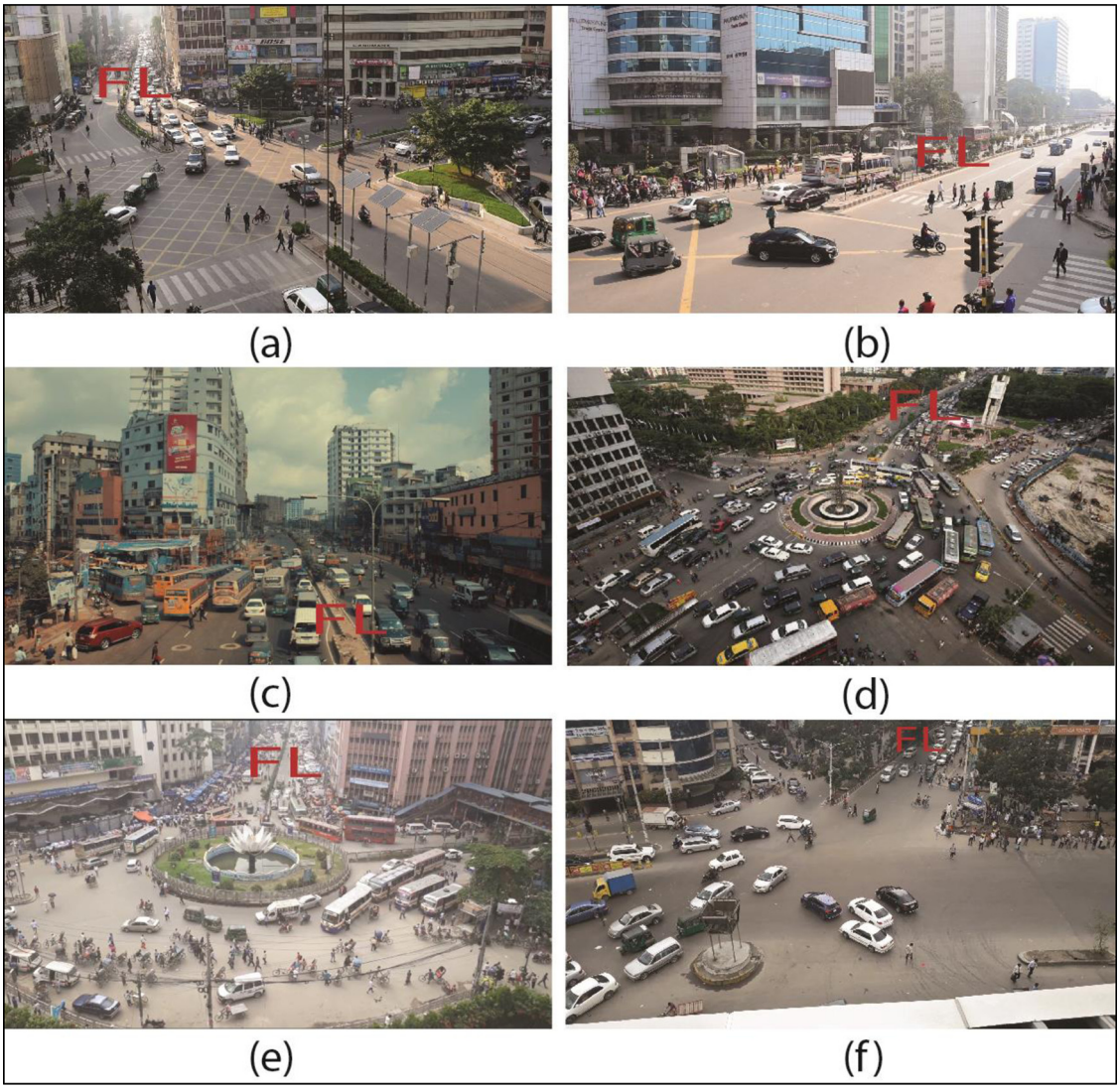


Fig. 2. Photographs of the intersections: (a) Gulshan-2, (b) Bangla Motor, (c) Shymoli, (d) Sonargaon-Panthapth-ETV, (e) Shapla Chattar, and (f) Jasin Uddin. FL means the focused leg of the intersection from where video footages were collected by placing a video camera closely for extracting necessary data.

Screenshots were taken from the video footages at ten-minute intervals. All the visible pedestrians in the screenshots who crossed the entire road (both near-side and far-side of the road) at intersection were taken as sample. It was done to avoid biasness in selecting sample pedestrians. In addition, if pedestrians, presented in the screenshots, crossed the road in group, then only one pedestrian was selected randomly as a sample from that group. In this way, 546 pedestrian samples were selected from the screenshots and required data of the pedestrian samples were observed from the video footages of the studied intersections.

Data of vehicle flow and available gap were extracted from the video footages. Vehicle flow was defined as the numbers of vehicles were passing on the focused leg of the intersection during a specific time period. The time period was measured from when the pedestrian expressed his/her intention to cross the road until the pedestrian finished his/her crossing. Available gap was defined as the gap available on the focused leg when the pedestrian was crossing the focused leg. This factor was measured in terms of time gap (seconds) between pedestrian and the nearest vehicle while crossing. Extraction procedure of this factor is presented with an example. Suppose, the pedestrian presents in the Fig. 3 was the sample pedestrian. At the time of crossing, this sample pedestrian interacted with three vehicles: *A*, *B*, and *C*. Gaps between the sample pedestrian and *A*, *B*, and *C* vehicles were *x* (Fig. 3-a), *y* (Fig. 3-b), and *z* (Fig. 3-c), respectively. Then, the times required by *A* vehicle to travel *x* distance, *B* vehicle to travel *y* distance, and *C* vehicle to travel *z* distance were extracted from the video footages. For simplification, *A* vehicle needed *i* seconds to travel *x* distance; whereas, *B* and *C* vehicles needed *j* and *k* seconds to travel *y* and *z* distance, respectively. The lowest value among the *i*, *j*, and *k* was identified as the value of available gap. Apart from this, if a sample pedestrian did not interact with any vehicle while crossing (vehicle flow was zero) then the available gap was categorized as *large gap* (Table 2).

Though data related to gender of the sample pedestrian were easily extracted from the video footages, it was difficult to extract the data of age. Approximate age of sample pedestrian was estimated from their visual physical appearance and extracted from the video footages.

Pedestrian road-crossing behavior (road crossing strategy, crossing group size, and using crosswalk) related data were also extracted from the video footages. Pedestrian crossing speed was estimated through dividing the crossing distance (focused leg width) by required crossing time (waiting time of the pedestrian was not included). Waiting time was defined as the time pedestrian waited for getting crossing opportunity. It was estimated by adding the waiting time at starting location and waiting time at median. Waiting time at starting location was the time between pedestrian expressed his/her intention to cross at starting location and pedestrian started to cross. Waiting time at median was the time between pedestrian reached to the median by crossing the near-

Table 2
Details about collected data.

Variable	Types of variable	Definition and Label
Intersection geometry and traffic environmental characteristics		
Intersection control type	Categorical	*Controlled:* Intersection was controlled by traffic police manually or implementing the direction of vehicular traffic signals. *Uncontrolled:* Intersection was neither controlled by traffic police nor traffic light.
Crosswalk condition	Categorical	*Highly visible:* Crosswalk was present in the focused leg and crosswalk marking was not fully or partially eroded. *Poorly visible:* Crosswalk was present in the focused leg and crosswalk marking was partially eroded. *Not available:* Crosswalk was not present in the focused leg or crosswalk marking was fully eroded.
Road width	Continuous	Width of the focused leg in meters
Median width	Continuous	Width of the median of the focused leg in meters
Vehicle flow	Continuous	Number of vehicles were passing on the focused leg of the intersection during the time period of sample pedestrian crossing. It was expressed in vehicle/ minute unit.
Available gap	Categorical	Gap available on the focused leg when the pedestrian was crossing the focused leg. It was categorized into three levels. *Small gap:* Time gap between pedestrian and the nearest vehicle was less than 4 s. *Medium gap:* Time gap between pedestrian and the nearest vehicle was 4–7 seconds. *Large gap:* Time gap between pedestrian and the nearest vehicle was greater than 7 s.
Pedestrian characteristics		
Gender	Categorical	Gender of pedestrian: *Male* and *Female*
Age	Categorical	Estimated age of pedestrian from visual appearance: *Child*: 0–15, *Young*: 15–35, *Adult*: 35 − 60, and *Old*: above 60 years
Pedestrian road-crossing behavior		
Road-crossing strategy	Categorical	*Rolling gap*: Pedestrians discontinuously crossed the road by searching gaps between the continuous flow of vehicles by adjusting their speed and direction of movement, and consequently they rolled over the small vehicular gaps to cross the road (Fig. 1). *Other*: Pedestrian who did not cross by rolling gap crossing.
Crossing group size	Categorical	*Alone:* Pedestrian singly crossed the focused leg. *Pair:* Two pedestrians crossed the focused leg in a group. *Group:* More than two pedestrians crossed the focused leg in a group.
Using crosswalk	Categorical	*No:* Pedestrian did not use crosswalk while crossing. *Yes:* Pedestrian used crosswalk while crossing.
Crossing speed	Continuous	Crossing speed of the pedestrian in meter/ second unit
Waiting time	Continuous	Waiting time of the pedestrian at starting location and median location in seconds

side of the road and started to traverse to cross the far-side of the road. With all the collected data, a database was created in Statistical Package for the Social Sciences (SPSS) for examining pedestrian road-crossing strategy.

3.4. Data analysis

In order to fulfill the aim of the study, logistic regression methodology was applied and completed in two stages. In the first stage, univariate analysis was conducted to examine the relationship between each factor with crossing strategy separately. In univariate analysis, descriptive analysis and statistical analysis were conducted. To examine the relationship between categorical factors with crossing strategy through statistical analysis, Chi-Square test was conducted; in contrast, Independent-Sample T test was conducted for continuous factors. In the second stage, multivariate analysis was conducted by developing a binary logistic regression model. To develop this model, factors having a p-value of less than 0.25 in the univariate analysis were selected as independent variables. According to the Hosmer and Lemeshow (2000), any factor having a p-value less than 0.25 in univariate test should be an independent variable for the multivariate model. Presence of multicollinearity among the independent variables was checked by the Variance Inflation Factor (VIF) test. Crossing strategy (the likelihood of crossing the road by rolling gap crossing strategy) was the dependent variable. Using the dependent and independent variables, a binary logistic regression model was developed in SPSS using the stepwise selection method. Independent variables that were found to be significant at 95% confidence level in the developed model were considered as significant variables. After developing the model, the fitness of the model with data was checked with the help of model summary statistics (Chi-Square test) and Hosmer and Lemeshow test.

4. Result and discussion

4.1. Univariate analysis

To identify factors that influenced pedestrians' decision to cross the road by rolling gap crossing, 12 potential factors were considered in this study. Among them, seven factors were categorical and the rest of the five factors were continuous. Descriptive statistics and relationship of twelve potential factors with pedestrian crossing strategy are shown in Table 3 and Table 4. As presented in Table 3, the study sample consisted of approximately 31% rolling gap crossing (n = 169) and 69% other crossing strategy (n = 377). The results of the descriptive analysis (Table 3) showed that the rate of exhibiting rolling gap crossing was higher at uncontrolled intersection (96.0%) than controlled intersections (16.4%). Pedestrians crossed the road by rolling gap crossing more where crosswalk was not available (64.3%) than those intersections where highly visible crosswalk was available (20.0%). The rate of exhibiting rolling gap crossing was lower when large gap was available on the road (03.8%) than when the available gap was smaller (43.4%). Moreover, the male pedestrian rate of showing rolling gap crossing (34.0%) was higher than the female pedestrian rate (22.4%). In addition, old pedestrians (17.9%) showed less rolling gap crossing than young (30.8%) and middle (35.2%) aged pedestrians. Pedestrians who crossed the road in pair (19.2%) showed less rolling gap crossing than pedestrians who crossed the road alone (31.6%). However, pedestrians who crossed the road in group (48.1%) showed more rolling gap crossing than those pedestrians who crossed the road alone (31.6%) or in a pair (19.2%). In addition, the rate of showing rolling gap crossing was found to be lower for the pedestrians who crossed the road by using crosswalk (13.7%) than those who did not use crosswalk (60.4%).

In Table 4, the mean and standard deviation value of the continuous factors are presented. From the table, it can be said that pedestrians who crossed the road slowly showed more rolling gap crossing strategy. On the other hand, pedestrians who waited a long time showed less rolling gap crossing. In the case of road width, pedestrians showed more rolling gap crossing strategy in wider road. On the contrary, pedestrians showed less rolling gap crossing where wider median was present. Finally, it was also found that pedestrians showed higher rolling gap crossing when vehicle flow was higher.

For statistical analysis, Chi-Square test was conducted for categorical factors (Table 3) and Independent-Samples T test was conducted for continuous factors (Table 4). All the factors except waiting time of pedestrian were found statistically significant at the 95% confidence level (p- value < 0.05). So, it can be said that these 11 statistically significant factors had a strong association with pedestrian crossing

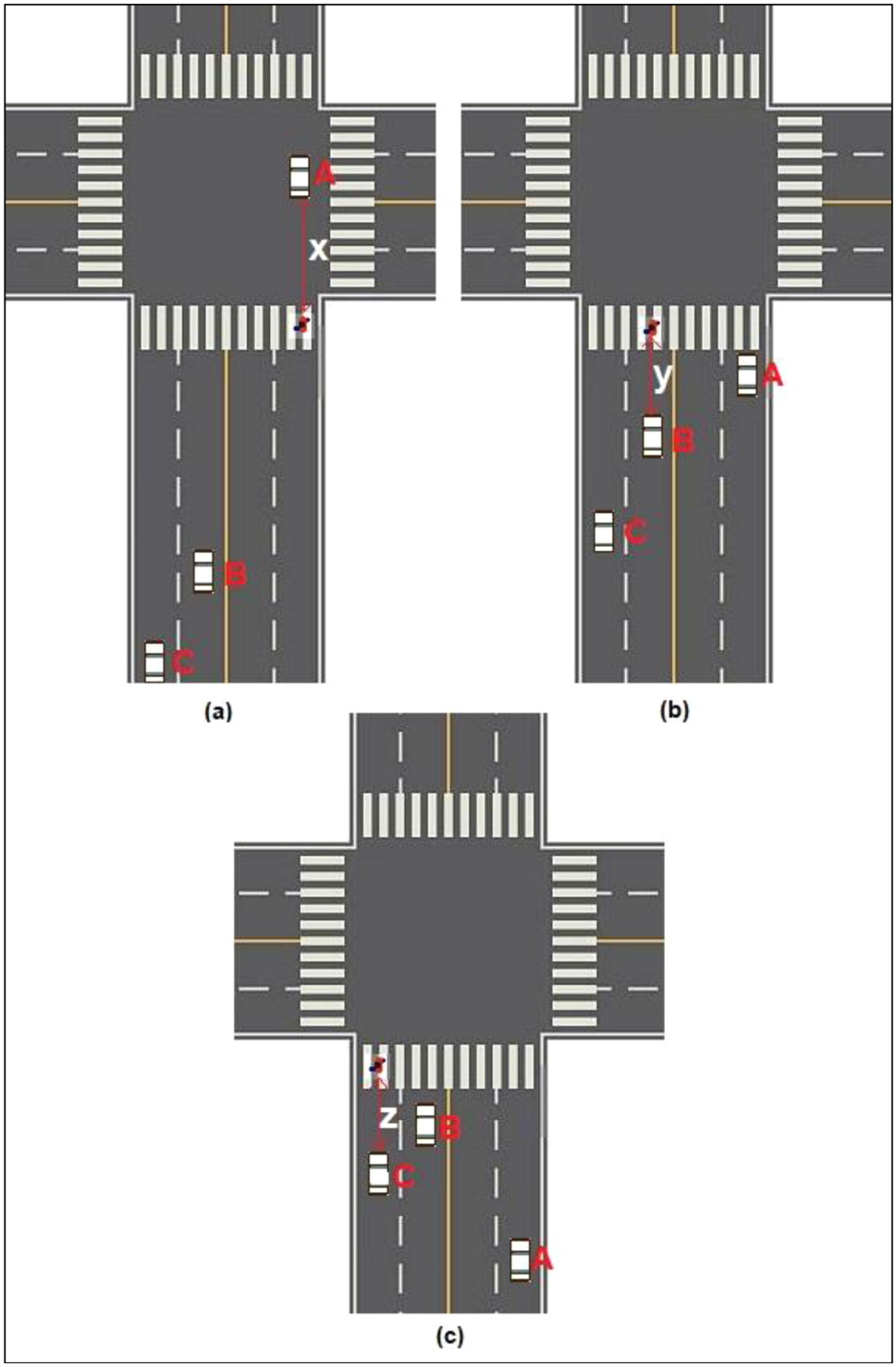


Fig. 3. Concept of estimating available gap on the road.

strategy and influenced pedestrians to decide whether they crossed the road by rolling gap or other crossing strategies.

4.2. Multivariate analysis

In the multivariate analysis, a binary logistic regression model was developed. To estimate the likelihood of showing rolling gap crossing by pedestrian, first, it was required to select the factors that should be considered as independent variables for developing the model. The statistical analysis (Section 4.1) results showed that all the factors except waiting time of pedestrian had a significant effect on the likelihood of showing rolling gap crossing (p- value < 0.05). For the waiting time of pedestrian factor, p-value was found to be 0.117. According to the Hosmer and Lemeshow (2000), any factor having a p-value less than 0.25 in the univariate test should be an independent variable for the multivariate model. So, all the twelve factors were considered as independent variables for developing a binary logistic regression model. Then, multicollinearity among the independent variables was checked using the VIF test. No multicollinearity issue was found among the independent variables as all the VIF values were found

Table 3
Descriptive statistics and association of categorical factors with crossing strategy.

Variable	Crossing Strategy		Total [*n (%)*]	χ^2 (*df*) [*p*-value]
	Rolling gap [*n(%)*]	Other [*n (%)*]		
Intersection control type				
Controlled	73 (16.4%)	373 (83.6%)	446 (81.7%)	242.3(1)
Uncontrolled	96 (96.0%)	004 (04.0%)	100 (18.3%)	[0.000]
Crosswalk Condition				
Highly visible	36 (20.0%)	144 (80.0%)	180 (33.0%)	152.15(2)
Poorly visible	14 (7.7%)	167 (92.3%)	181 (33.2%)	[0.000]
Not available	119 (64.3%)	66 (35.7%)	185 (33.9%)	
Available gap				
Small	147 (43.4%)	192 (56.6%)	339 (62.1%)	69.07 (2)
Medium	18 (17.6%)	84 (82.4%)	102 (18.7%)	[0.000]
Large	4 (3.8%)	101 (96.2%)	105 (19.2%)	
Gender				
Female	32 (22.4%)	111 (77.6%)	143 (26.2%)	6.66 (1)
Male	137 (34.0%)	266 (66.0%)	403 (73.8%)	[0.011]
Age				
Children	00 (0.0%)	00 (0.0%)	00 (0.0%)	6.00 (2)
Young	95 (30.8%)	213 (69.2%)	308 (56.4%)	[0.50]
Middle	64 (35.2%)	118 (64.8%)	182 (33.3%)	
Old	10 (17.9%)	46 (82.1%)	56 (10.3%)	
Crossing group size				
Alone	124 (31.6%)	269 (68.4%)	393 (72%)	13.94 (2)
Pair	19 (19.2%)	80 (80.8%)	99 (18.1%)	[0.001]
Group	26 (48.1%)	28 (51.9%)	54 (9.9%)	
Using crosswalk				
No	122 (60.4%)	80 (39.6%)	202 (37.0%)	130.05 (1)
Yes	47 (13.7%)	297 (86.3%)	344 (63.0%)	[0.000]
Total	169 (31.0%)	377 (69.0%)	546 (100%)	

to be less than 2. Finally, these independent variables were inputted into the model using the stepwise selection method (selection method = forward: conditional) of LOGISTIC REGRESSION in SPSS. Results of the developed model are displayed in Table 5 that includes the coefficients, odds ratio and its 95% confidence interval (*CI*), and *p*-values for each independent variable, as well as also includes model summary statistics.

The model summary statistics showed the value of Chi-square statistics ($\chi 2$), degree of freedom (*df*), and *p*-value of the developed model (Table 5). These values showed that this model was found to be statistically significant at the 99% confidence level (*p*-value < 0.01). Nagelkerke R^2 was found 0.663 indicating that this model explained 66.3% of the variance in the likelihood of showing rolling gap crossing by pedestrian. This model correctly classified 87.0% of cases. Furthermore, the Hosmer–Lemeshow statistic was not found to be statistically significant for this model (*p*-value > 0.05). All the model summary statistics and the result of Hosmer and Lemeshow test confirmed that the developed model fitted the data well.

From Table 5, it can be said that seven variables among the twelve independent variables were found to be statistically significant in the developed model (p-value < 0.05). The likelihood of showing rolling gap crossing by pedestrian had been influenced by these seven significant variables: intersection control type, available gap, median width, vehicle flow, age group, crossing group size, and using crosswalk.

Table 5
Results of the stepwise binary logistic regression model.

Variable	Coef. (*B*)	Odd ratio (*OR*)	95 % *CI*		*p*-value
			Lower	Upper	
Constant	1.787	5.969			0.031**
Intersection control type (Ref: uncontrolled)					
Controlled	−3.111	0.045	0.013	0.158	0.000*
Available gap (Ref: small)					
Large	−1.377	0.252	0.084	0.760	0.014**
Median width	−1.129	0.323	0.183	0.573	0.000*
Vehicle flow	0.021	1.021	1.013	1.029	0.000*
Age group (Ref: young)					
Old	−1.681	0.186	0.056	0.621	0.006*
Crossing group size (ref: pair)					
Alone	1.273	3.571	1.462	8.722	0.005*
Using crosswalk (Ref: no)					
Yes	−0.720	0.487	0.258	0.920	0.027**

Model summary.
n = 546, χ^2 = 347.485, *df* = 07, *p*-value = 0.000, -2 Log likelihood = 328.156, Nagelkerke *R*-square = 0.663, Correctly classified = .87.0%.
Hosmer and Lemeshow test.
χ^2 = 14.203, *df* = 08, *p*-value = 0.077.
* indicates significance at the 1% level and ** at the 5% level.

In general, for a continuous independent variable, a positive coefficient (*B*) value means that the likelihood of showing rolling gap crossing increases for an increase in the value of that variable; for a categorical independent variable, it means that maintaining the corresponding particular category increases the likelihood of showing rolling gap crossing compared to the reference category. Besides, for a negative coefficient (*B*) value means vice versa. Odd ratio (*OR*) statistics is also used to interpret regression model easily (*OR* is the exponentiation of coefficients). For a continuous variable, if the *OR* value *X* is greater than one, then it means that the likelihood of showing rolling gap crossing increases *X* times for per unit increase of that variable. Besides, for a categorical variable, if the *OR* value *X* is greater than one for a particular category, then it means that the likelihood of showing rolling gap crossing increases *X* times of that particular category as compared to the reference category. On the other hand, for an odd ratio of less than one means vice versa. Discussion on the model results is given in the following sections.

4.2.1. Intersection geometry and traffic environmental characteristics

Intersection control type was found one of the most influential significant factors in this study. The likelihood of showing rolling gap crossing by pedestrian was almost 5.9 times higher at uncontrolled intersection than controlled intersection, holding other independent variables constant. This result indicated that pedestrians showed risky behavior more at uncontrolled intersection. At controlled intersection, pedestrians could cross the road at protected phase if they wish; thus, they could cross the road when no vehicle moved on the road, and consequently they did not need to roll over the vehicular gap. At busy

Table 4
Descriptive statistics and correlation of continuous factors with crossing strategy.

Variable	Crossing Strategy		Total [Mean (S.D.)]	*t* (*df*) [*p*-value]
	Rolling gap [Mean (S.D.)]	Other [Mean (S.D.)]		
Crossing speed of pedestrian	1.14 (0.3)	1.26 (0.4)	1.22 (0.38)	3.6 (400.7) [0.000]
Waiting time of pedestrian	13.4 (18.6)	16.5 (26.5)	15.5 (24.3)	1.5 (445.8) [0.117]
Road width	29.6 (2.2)	26.3 (2.0)	27.3 (2.6)	16.4 (544) [0.000]
Median width	1.3 (0.2)	1.8 (0.6)	1.69 (0.6)	13.5 (534.3) [0.000]
Vehicle flow	94.7 (34.1)	46.6 (33.5)	61.5 (40.3)	−15.4 (544) [0.000]

uncontrolled intersection, vehicle flow was generally found continuous, and no protected phase was available for pedestrians to cross the road without interacting with the vehicle. As a result, pedestrians were forced to cross the road by seeking gaps between the vehicles. It could be a likely explanation why pedestrians showed more risky rolling gap crossing at uncontrolled intersection than controlled intersection. Previous studies demonstrated that pedestrians showed less risky behavior at intersections having better (or advance) traffic control system (Brosseau et al., 2013; Asaithambi et al., 2016; zafri et al., 2019). This finding was found consistent with the result of this study.

The likelihood of showing rolling gap crossing was found to be 0.25 times lower when large gap was available on the road compared to the situation when small gap was available, holding other independent variables constant. When pedestrians got large gap on the road, they could cross the road directly, and did not need to roll over the small vehicular gap; therefore, pedestrians showed less rolling gap crossing when large vehicular gap was available on the road. Brewer et al. (2006); Kadali and Vedagiri (2013), and Chandra et al. (2014) studied on pedestrian gap acceptance behavior and they also showed a similar result in their studies. They showed that pedestrians might cross the road by rolling gap crossing when they did not find appropriate gap in traffic on the road.

Median width was also found to be a significant factor. The likelihood of showing rolling gap crossing type decreased 0.32 times for one-unit increase in median width, holding other independent variables constant. An explanation could be that if a median was wide enough, a pedestrian could wait on the median for a long time to get the far-side of the road to be cleared. For that, they did not need to interact with the vehicle while crossing. Therefore, a large median might influence the pedestrians to cross the road in two-stage crossing, which resulted in decreasing the exhibition of rolling gap crossing.

In the case of vehicle flow, it was found that the likelihood of showing rolling gap crossing increased 1.02 times for one-unit increase of vehicle flow, holding other independent variables constant. Increasing the number of vehicles on the road restricted the opportunity to cross the road directly. It forced pedestrians to adjust their speed and direction of crossing with vehicle flow, and to roll over the small vehicular gap to cross the road. Therefore, pedestrians need to cross the road by rolling gap crossing when vehicle flow was found higher. A similar result was found in the study of Zhang et al. (2018).

4.2.2. Pedestrian characteristics

Among the pedestrian characteristics, pedestrian age group was found as the most influential significant factor. Old pedestrians showed 0.18 times less rolling gap crossing than young pedestrians. Generally, young pedestrians took more risk and showed more risky behavior at the time of crossing than old pedestrians (Granié et al., 2013; Jain et al., 2014; Ravishankar and Nair, 2018). Young pedestrians did not want to wait more for the road to be cleared (Guo et al., 2011). In addition, they violated traffic signals more than old pedestrians, and consequently crossed the road when vehicle flow was continuous (Granié et al., 2013; Ren et al., 2011; Zafri et al., 2019). As a result, young pedestrians showed rolling gap crossing more compared to the old pedestrians.

4.2.3. Pedestrian road-crossing behavior

The likelihood of showing rolling gap crossing was found 3.5 times higher if a pedestrian crossed the road alone compared to the scenario when pedestrians crossed the road in pair, holding other variables constant. One of the likely explanation could be that when a pedestrian crossed the road alone, he/she had more flexibility to adjust his/her speed and direction of crossing, and rolled over the small vehicular gap; therefore, pedestrian who crossed the road alone showed more risky rolling gap crossing. Previous studies found that pedestrians showed risky behaviors more while crossing the road alone (Rosenbloom, 2009; Marisamynathan and Perumal, 2014; Brosseau et al., 2013; Yannis et al., 2013). This finding was found consistent with the result of this study.

It was also found that pedestrians who followed the crosswalk while crossing showed 0.48 times less rolling gap crossing at intersections compared to the pedestrian who did not follow, holding other variables constant. Drivers provided more priority and yielded more to the pedestrians who crossed the road by following crosswalk than those who did not follow (Balk et al., 2014). So, it was easier for the pedestrian who crossed the road by using crosswalk to cross directly by single/ two-stage crossing.

4.3. Limitation and future work

Two limitations of this study inherited from the method adopted for data collection. First, the accuracy of data collected by videography survey is good but not promisingly great. For example, age data of the pedestrians were collected from the video footages based on their physical appearance. These approximately estimated age data of the pedestrians might help to build up an understanding about their crossing strategy; however, the exact age of the individual pedestrian needs to be considered to improve the model performance. Second, data related to pedestrian psychology, education level, mental process, attitude, and perception about situation, safety, and risk were not possible to collect by videography survey method. Furthermore, pedestrians' decision related to their crossing strategy could also be influenced by other factors, including decision of the nearby pedestrians, driver yielding behavior, further variation in roadway characteristics (one-way/two-way, highway/ arterial road/ feeder road), availability of traffic signs and markings, surrounding land use, and so on. Future studies should be focused to identify the impact of the mentioned factors on pedestrian road-crossing strategy.

5. Conclusion and recommendations

The aim of this study was to explore risky rolling gap crossing strategy of the pedestrian and identify the factors that influenced the pedestrian to cross the road by rolling gap crossing strategy at intersection. To fulfill the aim, twelve factors related to intersection geometry, traffic environmental characteristics, pedestrian characteristics, and their road-crossing behaviors were considered. A binary logistic regression model was developed using these twelve factors to estimate the likelihood of showing rolling gap crossing by pedestrian. Seven factors were found to be statistically significant in the developed model: intersection control type, available gap, median width, vehicle flow, age group, crossing group size, and using crosswalk.

Results of the study showed that pedestrians showed more rolling gap crossing at uncontrolled intersection than controlled intersection. Therefore, uncontrolled intersection is suggested to be turned into controlled intersection by installing traffic signals, signs, and road markings. This study also indicated that pedestrians showed less rolling gap crossing on the roads having a wider median. So, the authority is advised to increase the width of the median according to the appropriate standard to reduce this risky behavior. Raising awareness among pedestrians could be useful to encourage pedestrians to avoid rolling gap crossing. This study showed that pedestrian showed lower rolling gap crossing when large gap was available on the road compared to the situation when smaller gap was available. In addition, the exhibition of rolling gap crossing increased when vehicle flow increased on the road. Thus, the pedestrians are advised to cross the road during the protected phase at controlled intersection. In uncontrolled intersection, pedestrians are advised to cross when large gap and fewer vehicles are available on the road. This study also showed that old pedestrians showed less rolling gap crossing than young pedestrians. Therefore, in awareness-raising programs, special attention need to be given on young pedestrians to encourage them to avoid rolling gap crossing. Rolling gap crossing was found to be higher if a pedestrian crossed the road alone rather than in pair. Therefore, pedestrians are suggested not

to cross the road alone. Lastly, this study showed that pedestrians who followed the crosswalk while crossing showed lower rolling gap crossing at intersection compared to the pedestrian who did not follow the crosswalk. So, pedestrians are advised to use crosswalk while crossing. Highly visible crosswalk is recommended to be provided in all the crossing locations, as well as drivers and pedestrians are suggested to be educated about the use of crosswalk. The outcomes of the study would be useful for the policymakers to take necessary interventions to reduce pedestrian safety-related problems; therefore, making a more walkable and safe city for the pedestrians.

CRediT authorship contribution statement

Niaz Mahmud Zafri: Conceptualization, Formal analysis, Investigation, Methodology, Software, Writing - original draft, Writing - review & editing. **Rashada Sultana:** Conceptualization, Formal analysis, Methodology, Writing - original draft, Writing - review & editing. **Md. Rakibul Hasan Himal:** Conceptualization, Formal analysis, Methodology, Writing - original draft, Writing - review & editing. **Tanzila Tabassum:** Conceptualization, Formal analysis, Methodology, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.aap.2020.105564.

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